

Prague University of Economics and Business
Faculty of Informatics and Statistics



Finite Sample Properties of Generalized Autoregressive Score Models

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Author: Bc. Jan Švrčina

Supervisor: Ing. Petra Tomanová, MSc., Ph.D.

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Abstrakt

Tato práce se zabývá vlastnostmi GAS (Generalized Autoregressive Score) modelů. Je v ní teoretický základ pro oblast analýzy časových řad a také popis GAS modelů a jejich vlastností. Také je ukázáno jak GAS modely souvisí s již dříve prezentovaným modely s časově proměnlivými parametry, jako jsou GARCH, ADC a ADI modely. Dále také prezentuje simulační studii o 29700 vzorcích z odhadnutých GAS modelů na simulačních datech, za použití R knihovny *gasmodel*, a to přes velký rozsah pravěpodobnostních distribucí i parametrů modelu. Tato studie empiricky ukazuje konzistenci a přesnost rostoucí s velikostí výběru některých prezentovaných GAS modelů a případy kde modely nekonvergovaly jsou popsány a je vedena diskuze z jakých důvodů se tak stalo.

Klíčová slova

GAS modely, modely s časově proměnlivými parametry, analýza časových řad, simulační studie

Abstract

This thesis discusses the finite sample properties of generalized autoregressive score models (GAS). It provides a theoretical background to the area of time-varying parameter models as well as a description of GAS model and its properties. It shows the connection between GAS models and previously developed time-varying parameter models such as GARCH, ADC, and ADI models. Furthermore, it presents a simulation study of 29700 samples for the estimated GAS models on simulated data, using the R package *gasmodel*, across various probability distributions and model parameters. The simulation study results then empirically show the consistency and precision increasing with the sample size of some of the presented GAS models. The cases where the models have not converged are outlined, and a possible explanation is discussed.

Keywords

generalized autoregressive score models, time varying parameter models, times series analysis, simulation study

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List of abbreviations

ACD Autoregressive Conditional Duration	GARCH Generalized Autoregressive Conditional Heteroskedasticity
ACI Autoregressive Conditional Intensity	GAS Generalized Autoregressive Score
AR Autoregressive	MA Moving Average
ARMA Autoregressive Moving Average	MLE Maximum Likelihood Estimate
ARMAX Autoregressive–moving-average model with Exogenous inputs	MSE Mean Square Error
CAViaR Conditional Autoregressive Value at Risk	LIE Law of Iterated Expectations
CLT Central Limit Theorem	OoB Out of Bounds
DCC Dynamic Conditional Correlation	PDF Probability Density Function

1. Introduction

1.1 Introduction to GAS models

When it comes to the analysis of time series processes, often, the models based on static parameters are unable to capture the dynamic relations that are driving these processes, as such models with dynamic parameters, parameters that vary in time, have to be introduced to explain the time series process adequately. These models have been categorized and described in "Statistical Analysis of Time Series: Some Recent Developments" by D. R. Cox (1).

Generalized autoregressive score (GAS) models (2) are a class of observation-driven models that is a generalization of previously introduced models, such as generalized autoregressive conditional heteroskedasticity (GARCH) (3), autoregressive conditional duration and intensity (ACD and ACI) (4) (5), dynamic conditional correlation (DCC) (6) and other models further described in this thesis.

— In the observation-driven approach, the time variation of the parameters is introduced by letting parameters be functions of lagged dependent variables as well as contemporaneous and lagged exogenous variables. Although the parameters are stochastic, they are perfectly predictable, given the past information. This approach simplifies likelihood evaluation and explains why observation-driven models have become popular in the applied statistics and econometrics literature. — "Generalized Autoregressive Models with Applications" by D. Creal, S. J. Koopman and A. Lucas (2)

the core of the GAS model is the updating mechanism for the time-varying parameters $\mathbf{f}_t$ specified as follows (2)

$$\mathbf{f}_{t+1} = \boldsymbol{\omega} + \sum_{i=1}^q \boldsymbol{\alpha}_i s_{t-i+1} + \sum_{i=1}^p \boldsymbol{\varphi}_i \mathbf{f}_{t-i+1}, \quad (1.1)$$

where q and p are the score order and autoregressive order of the model, $\boldsymbol{\omega}$ is a vector of constants, $\boldsymbol{\alpha}_i$ for $i \in \{1, \dots, q\}$ are matrices of score coefficients, $\boldsymbol{\varphi}_i$ for $i \in \{1, \dots, p\}$ are matrices of autoregressive coefficients. All three of $\boldsymbol{\omega} = \boldsymbol{\omega}(\boldsymbol{\theta})$, $\boldsymbol{\alpha}_i = \boldsymbol{\alpha}_i(\boldsymbol{\theta})$, $\boldsymbol{\varphi}_i = \boldsymbol{\varphi}_i(\boldsymbol{\theta})$ are based on a vector of static parameters of the model $\boldsymbol{\theta}$. The function of the past data (score) $s_t = s(\mathbf{f}_t, \mathbf{y}_t, \mathbf{x}_t, \boldsymbol{\theta})$ is based on the time-varying parameters $\mathbf{f}_t$, observations $\mathbf{y}_t$, exogenous variables $\mathbf{x}_t$ and the static parameters $\boldsymbol{\theta}$.

Since the observed variables $\mathbf{y}_t$ are drawn from the distribution $p(\mathbf{y}_t | \mathbf{f}_t, \mathbf{x}_t; \boldsymbol{\theta})$ ¹

$$\mathbf{y}_t \sim p(\mathbf{y}_t | \mathbf{f}_t, \mathbf{x}_t; \boldsymbol{\theta}), \quad (1.2)$$

the score $\mathbf{s}_t$ is then composed of the steepest ascend direction of the parameters ∇_t for improving the log-observation density at time t scaled by a scaling function $\mathbf{S}_t(\cdot)$

$$\mathbf{s}_t = \mathbf{S}_t \nabla_t, \quad \nabla_t = \frac{\partial \ln p(\mathbf{y}_t | \mathbf{f}_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial \mathbf{f}_t}. \quad (1.3)$$

the most common choices for the scaling functions are identity $\mathbf{S}_t = \mathbf{I}$, (pseudo)inverse of the information matrix $\mathbf{S}_t = \mathcal{I}_t^{-1}$ or square root of the (pseudo)inverse of the information of matrix $\mathbf{S}_t = \mathcal{I}_t^{-1/2}$. The information matrix can also be interpreted as the variance of the unscaled score ∇_t

$$\mathcal{I}_t = \mathbb{E} \left[\nabla_t \nabla_t^T \right], \quad (1.4)$$

with $\mathbb{E}[\cdot]$ being the expected value, with respect to $p(\mathbf{y}_t | \mathbf{f}_t, \mathbf{x}_t; \boldsymbol{\theta})$ (2).

1.2 Motivation and goals

1.2.1 Motivation

GAS models are an active area of research in the field of econometrics, from both theoretical and empirical standpoint; there are more than 280 article references on the site <http://www.gasmodel.com>. Their flexibility (the probability distribution can be arbitrary) and the ability to interpret the estimated parameters and statistical properties makes them valuable tools for time series data analysis. With such an extensive framework arises the question of the limitations of this framework and how reliable these models are under limited or imperfect data.

1.2.2 Goals

the main goals of this thesis are to present the GAS framework along with its theoretical background in time series regression and to simulate and investigate the properties of the GAS

¹In the original article (2) the probability density function was introduced as dependent on all the previous time-varying parameters $\{\mathbf{f}_i\}_{i=1}^t$, observations $\{\mathbf{y}_i\}_{i=1}^{t-1}$, exogenous variables $\{\mathbf{x}_i\}_{i=1}^t$ as well as the current values and the static parameters. However, this definition would be redundant for the purposes of this thesis, and most models in practice use the introduced simplified version; because of this, the extended definition has been omitted.

model framework on finite sample data with varying distribution and model parameters. To achieve this goal, current implementations of GAS process simulations and GAS model estimators will be used to create a large sample of estimated models, and their performance will be analyzed.

1.3 Thesis structure

This thesis is divided into five chapters. The first chapter is an introduction, introducing the concept of GAS models and their specifications and outlining this thesis's motivation, goals, and structure. The second chapter is a literature review, listing relevant sources, starting with the overview of specific observation-driven models that inspired the GAS model framework, followed by sources introducing the GAS framework and the current state of the art of score-driven models. Furthermore, some interesting applications of current empirical uses of GAS models are presented. The third chapter is the theoretical background for understanding the GAS model and its connection to other models and properties. The fourth chapter is the methodology, discussing the technical detail regarding the simulation study. The fifth chapter presents selected results from the simulation study. The sixth chapter examines the presented results and tries to hypothesize an explanation for some of the unexpected results. The last chapter is the conclusion, reflecting on the thesis as a whole and evaluating the results.

2. Literature review

An excellent source for information regarding GAS models is the web <http://www.gasmodel.com>, which lists nearly 300 articles and has been a great inspiration for this chapter.

2.1 Overview of time-varying models

As mentioned, GAS models are directly connected to the previously introduced specific variants of time-varying models. Namely mentioned in the original paper "Generalized Autoregressive Models with Application" by D. Creal, S. J. Koopman, and A. Lucas in 2013 (2) are the following:

- GARCH model proposed in "Generalized Autoregressive Conditional Heteroskedasticity" (3) by T. Bollerslev in 1986 was a generalization of the ARCH model (7) which capitalized on the idea that current volatility (the time-varying parameter of the model) is not only based on the previous data but also the previous volatility thus being able to explain a broader range of heteroskedastic processes.
- ACD model proposed in "Autoregressive Conditional Duration: A New Model for Irregularly Spaced Transaction Data" (4) by R. F. Engle and J. R. Russell in 1998 as a corollary to the ARCH model to explain clusters of financial transactions.
- ACI model proposed in "Econometric Modeling of Multivariate Irregularly-Spaced High-Frequency Data" (5) by J. R. Russell in 2001 as an extension to model intensity (aggregate data from duration) with similar ideas to ADC models.
- DCC model proposed in "Dynamic Conditional Correlation: A Simple Class of Multivariate Generalized Autoregressive Conditional Heteroskedasticity Models" by R. Engle in 2002 as an extension of the univariate GARCH by introducing conditional parameters for correlations to tackle multivariate problems.
- "Observation driven models for Poisson counts" (8) by R. A. Davis in 2003 discussing the properties of observation-driven models whose explanatory variables follow a Poisson distribution.
- CAViaR model proposed in "CAViaR: Conditional Autoregressive Value at Risk by Regression Quantiles" (9) by R. F. Engle and S. Manganelli in 2004, which used the aggressive framework on the Value at Risk measure at specific quantiles to forecast market risk.

These were the directly connected resources regarding time-varying models; however, there are additional sources to be mentioned as relevant:

- "Statistical Analysis of Time Series: Some Recent Developments" (1) by D. R. Cox et. al. in 1981 differentiating between parameter-driven and observation-driven time-varying models and classifying them.

- The t-GARCH model proposed in "A Conditionally Heteroskedastic Time Series Model for Speculative Prices and Rates of Return" (10) by T. Bollerslev in 1987 extending the idea of GARCH with allowing t-distributions in the error terms, thus counteracting the effects of outliers with a heavier-tailed distribution.
- "Beta-t-(E)GARCH" (11) by A. Harvey and T. Chakravarty in 2008 introduced a parameter-driven model and thus further making the estimate of variance more robust to outliers.

2.2 Recent GAS extensions and applications

The following resources discuss the extensions or recent applications of GAS models:

- "Dynamic Tobit models" (12) by A. Harvey and Y. Liao extended the GAS framework to Tobit models and then forecasting data using GAS models even when the true model is parameter-driven.
- "Covid-19, credit risk management modeling, and government support" (13) by S. Telg, A. Dubinova and A. Lucas is an empirical study using empirical models to investigate rating and default risk dynamics over the covid-19 crisis.
- "Signal smoothing for score-driven models: a linear approach" (14) by S. Blazsek, A. Ayala and A. Licht extended the GAS model with linear data smoothing procedure and an application for US inflation rates is presented.
- "Forecasting extreme financial risk: A score-driven approach" (15) by F. Fuentes, R. Herrera, A. Clements introduced a model for forecasting extreme financial risk with respect to the duration and magnitude of these events.
- "Multivariate Markov-switching score-driven models: an application to the global crude oil market" (16) by S. Blazsek, A. Escibano and A. Licht introduces Markov switching score-driven model.

2.3 GAS and time-varying models performance

The following resources discuss the performance of GAS models or time-varying models generally:

- "Missing observations in observation-driven time series models" (17) by F. Blasques, P. Gorgi and S. J. Koopman discuss the impact of missing observations in observation-driven models and how to counteract the inconsistent inference of current methods.
- "Predicting Time-Varying Parameters with Parameter-Driven and Observation-Driven Models" (18) by S. J. Koopman, A. Lucas and M. Scharth in 2016 is a more recent review of time-varying parameter models, inspiring some benchmarking procedures in this thesis.

3. Theoretical background

This chapter discusses the general theoretical background, mainly focused on probability distributions and regression, necessary to understand the GAS model framework and its applications. It shows the relationship between the specific time-varying models and the GAS models.

3.1 Probability distributions

This section overviews probability distributions used in this thesis and their properties and motivation. The primary resource for this section was "Grinstead and Snell's Introduction to Probability" (19) by C. M. Grinstead and J. L. Snell, specifically chapter 5 Distributions and Densities.

To begin with, we start with a small note about notation. So far, we have used $p(\cdot)$ as a probability density function (PDF) for the continuous distributions and a probability mass function for the discrete variables. The probability mass function is simple, giving the probability that a random variable is equal to some value

$$p(x) = P[X = x]. \quad (3.1)$$

However, defining the probability density function in the same way would be wrong as for strictly continuous distributions, the probability of drawing an exact value is always zero; because of this we go for an integral definition of the PDF

$$\int_a^b p(x)dx = P[a \leq X \leq b]. \quad (3.2)$$

Further in this thesis, we will stick to the notation regarding probability mass functions and probability density functions for discrete and continuous distributions, respectively, the probability $P[\cdot]$ in terms of random variables is implied.

3.1.1 Negative binomial distribution

Negative binomial distribution (also known as Pascal distribution) (19, p 186-187) is a two-parameter discrete distribution. It is motivated by the question, "How many failures of simple Bernoulli trials (19, p 96) with a probability q until we see a specific number of successful trials k ". As such, we can define the probability mass function $p(x)$

$$p(x; k, q) = \binom{k+x-1}{k-1} q^k (1-q)^x, x \in \mathbb{N}_0, \quad (3.3)$$

we can explain this formula in the following way. We need any sequence of length $k+x-1$ and ending with a success, with $k-1$ successes anywhere and probability of q^k for the successes and $(1-q)^x$ for the failures. This distribution can be re-parameterized as mean μ and dispersion α ; this can be useful for estimating the distribution parameters from data.

$$\mu = k \frac{1-q}{q}, \alpha = \frac{1}{k}. \quad (3.4)$$

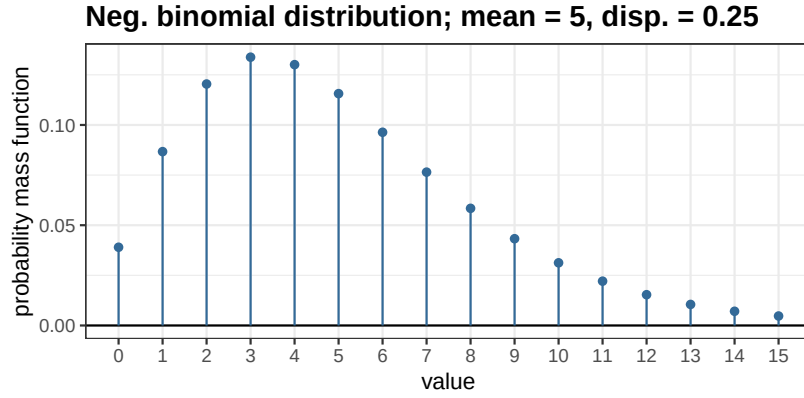


Figure 3.1: Probability mass function for the negative binomial distribution

3.1.2 Poisson distribution

Poisson distribution (19, p 187-192) is a one-parameter discrete distribution. It is motivated by the question, "How many occurrences of an event with a constant rate are we gonna get in a certain interval?". Its parameterized by a single value λ , which is also its mean and variance. We define the probability mass function

$$p(x; \lambda) = \frac{\lambda^x e^{-\lambda}}{x!}, x \in \mathbb{N}_0. \quad (3.5)$$

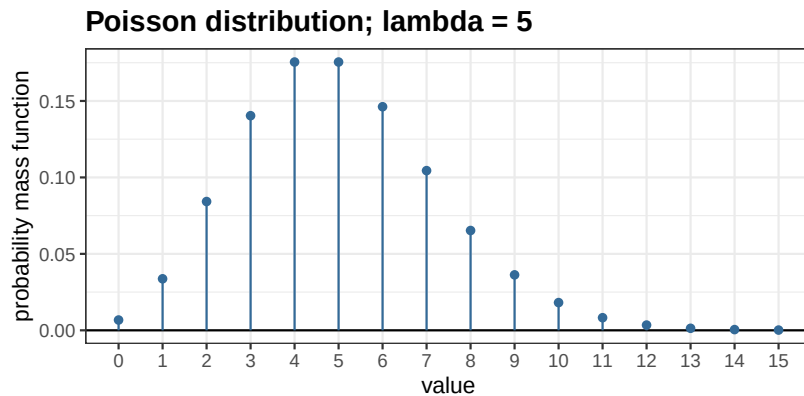


Figure 3.2: Probability mass function for the Poisson distribution

3.1.3 Normal distribution

Normal distribution (also known as Gaussian distribution) (19, p 212-215) is a two-parameter continuous distribution. Its importance arises from the central limit theorem (CLT) (19, ch. 9), as one of the key concepts in probability. It states that a sum of a large number of independent distributions converges to this normal distribution. This distribution is parameterized by two parameters, mean μ and variance σ^2 , the scale (standard deviation) is $\sqrt{\text{Var}} = \sigma$, define the probability density function as

$$p(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}. \quad (3.6)$$

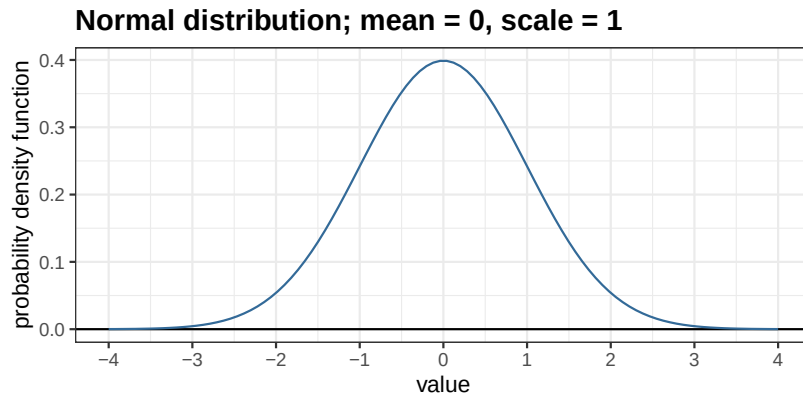


Figure 3.3: Probability density function for the normal (Gaussian) distribution

3.1.4 Student's t-distribution

Student's t-distribution (20)¹ is a generalization of the normal distribution with heavier tails (extreme values have higher probability density), as if the normal distribution describes the full population, the t-distribution describes a sample drawn from that population. This generalization is parameterized by parameter ν called degrees of freedom; with $\nu \rightarrow \infty$, the distribution is equivalent to the normal distribution. The probability density function is defined as

$$p(x; \mu, \sigma^2, \nu) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi\sigma^2}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{(x-\mu)^2}{\nu\sigma^2}\right)^{-\frac{\nu+1}{2}}, x \in \mathbb{R}. \quad (3.7)$$

In this definition is the gamma function Γ which is a generalization of the factorial function for the complex numbers (22)

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \text{Re}(z) > 0. \quad (3.8)$$

¹This distribution is named after the pseudonym *Student* of the original author of the distribution William Sealy Gosset (21).

In eq. 3.7 the distribution can then be transformed with parameter μ for mean and σ for scale similarly to the normal distribution, however unlike the normal distribution where $\text{Var} = \sigma^2$, for t-distribution $\text{Var} = \sigma^2 \frac{\nu}{\nu-2}$.

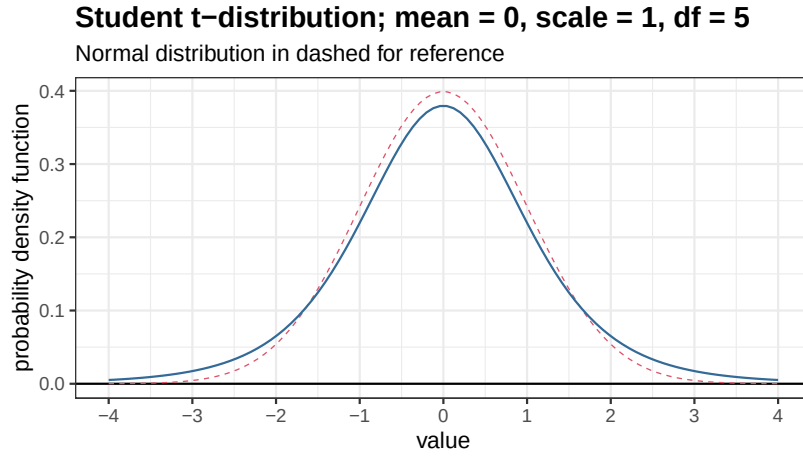


Figure 3.4: Probability density function for the Student t-distribution

3.1.5 Exponential distribution

Exponential distribution (19, p 205-209) is motivated by the question of "How long until the occurrence of an event with constant rate happens?". It is a 1-parameter continuous distribution; the parameter λ is the intensity of occurrence, $\frac{1}{\lambda}$ is the meantime. The definition of PDF is

$$p(x; \lambda) = \lambda e^{-\lambda x}, x \in \mathbb{R}_0^+. \quad (3.9)$$

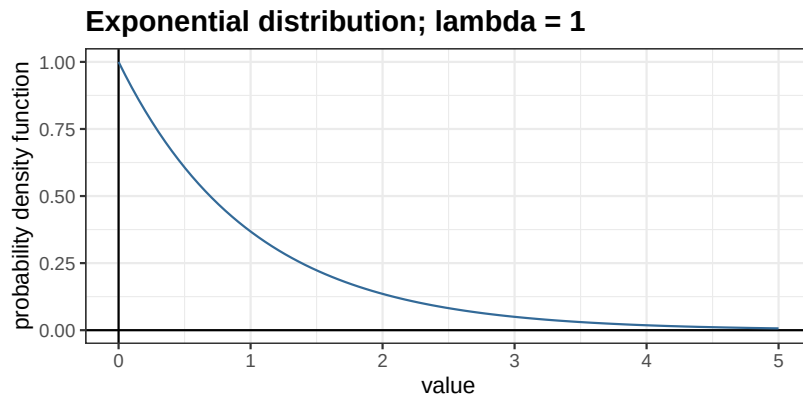


Figure 3.5: Probability density function for the exponential distribution

One of the important properties of this distribution is memorylessness of this distribution, that the condition that I have waited for some time t does not change the probability of how long I will still have to wait from now on

$$P[X > x] = P[X > x + t | X > t]. \quad (3.10)$$

To prove this, we will use the complementary cumulative $\bar{F}$

$$\bar{F}(x) = P[X > x] = \int_x^\infty p(x) = e^{-\lambda x}, \quad (3.11)$$

using conditional probability

$$P[X > x + t | X > t] = \frac{P[X > x + t \cap X > t]}{P[X > t]} \quad (3.12)$$

$$= \frac{P[X > x + t]}{P[X > t]} \quad (3.13)$$

$$= \frac{\bar{F}(x + t)}{\bar{F}(t)} \quad (3.14)$$

$$= \frac{e^{-\lambda(x+t)}}{e^{-\lambda t}} \quad (3.15)$$

$$= e^{-\lambda x} \quad (3.16)$$

$$= \bar{F}(x) \quad (3.17)$$

$$= P[X > x]. \quad (3.18)$$

3.1.6 Gamma distribution

Gamma distribution (19, p 208) is a two-parameter continuous distribution with varying uses, for example modeling waiting times in econometrics. It also has important special cases such as χ -squared distribution or the aforementioned exponential distribution. With parameterization of shape k and scale θ the PDF is defined as

$$p(x; k, \theta) = \frac{x^{k-1} e^{-\frac{x}{\theta}}}{\theta^k \Gamma(k)}, x \in \mathbb{R}_0^+, \quad (3.19)$$

with Γ as the gamma function defined in eq. 3.8. There is also an alternative parametrization with shape $\alpha = k$ and inverse scale (rate) $\beta = \frac{1}{\theta}$

$$p(x; \alpha, \beta) = \frac{x^{\alpha-1} e^{-\beta x} \beta^\alpha}{\Gamma(\alpha)}, x \in \mathbb{R}_0^+. \quad (3.20)$$

3.1.7 Weibull distribution

Weibull distribution (23) is a 2-parameter continuous distribution. Its main uses are in modeling time to failure or time between events, for example, how long a user spends on a web page. With parameterization of shape k and scale λ , the PDF is defined as

$$p(x; k, \theta) = \frac{k}{\theta} \left(\frac{x}{\theta} \right)^{k-1} e^{-\left(\frac{x}{\theta} \right)^k}, x \in \mathbb{R}_0^+, \quad (3.21)$$

as you can see, the exponential distribution is a special case for this distribution for $k = 1$, so Weibull distribution is a generalization of the exponential distribution; the intuition behind

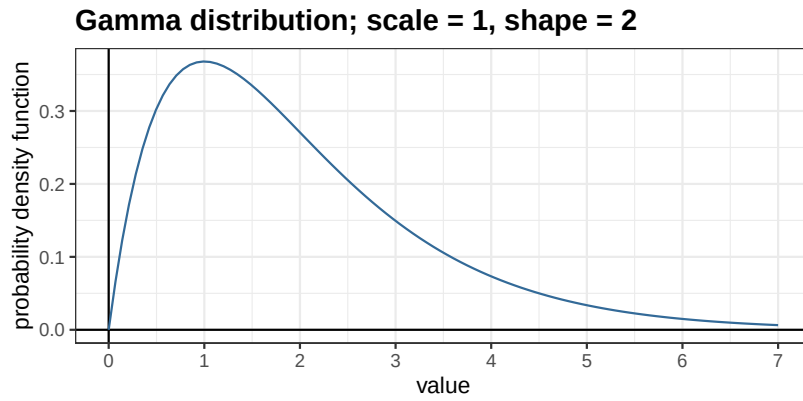


Figure 3.6: Probability density function for the gamma distribution

the shape parameter k is that for $k < 1$ the time between events gets shorter over time and for $k > 1$ the time between events increases over time.

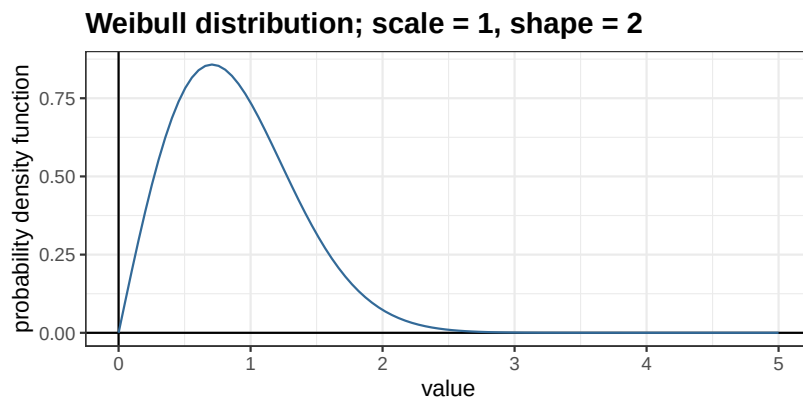


Figure 3.7: Probability density function for the Weibull distribution

3.2 Regression introduction

This section focuses on explaining the basics of regression and then using these stepping stones to arrive at the more complicated time series models, which inspired the GAS models. Most of these concepts are taken from the book "Introductory Econometrics" (24) by J. M. Wooldridge when regarding basics of regression and then "Time Series Analysis: Forecasting and Control" (25) by G.E.P. Box et. al. and "Time Series Analysis and Its Applications: With R Examples" by R.H. Shumway and D. S Stoffer when regarding the time-series analysis.

3.2.1 Motivation

Regression models are used to explain the relationship between variables. In a simple regression model, we are asking the question, "How do we explain the dependent variable y in terms

of the independent variable x ?", simply we can denote this causal relationship $x \rightarrow y$ (24, p 22).

If we had an exact value of $y \in Y$ for every possible $x \in X$, we would be able to define a function $f : X \rightarrow Y$ so $y = f(x)$, we would not need a regression model, in practice this would work only for discrete variables. Having uncountable number measurements is impossible.

Let's say we have exact values of y for only some of the values of x . In this case, we could use the interpolation/extrapolation techniques, for example, described in (26). The values produced by these techniques might not be correct but still assume that the relation is fixed without external factors.

Regression comes into play when the dependent variable y is not based only on some independent variable x but also some disturbance e , often described as the error term; it is the unobserved data, the causal relationship is then $x \rightarrow y \leftarrow e$. This concept is the motivation behind regression, and we are trying to make assumptions about the disturbance e and build a model for getting the best estimate of the relationship $x \rightarrow y$.

3.2.2 Dealing with the disturbance

Since the disturbance e is unobserved, we treat it like a random variable, transforming the functional relationships into probabilistic ones. If we have a random sample of data $\{x_i\}_{i=1}^N$, $\{y_i\}_{i=1}^N$ and we want to estimate the relationship $x \rightarrow y$, will try to find the probabilistic model of seeing y when we know x so we are searching for $p(y|x)$. If we talk about parametric regression, the probability function can be described using parameters θ (also known as the parameters of the regression model), so $p(y|x) = p(y|x; \theta)$, this conceptualization allows us to deal with the disturbance using tools from probability theory and to "fit" our model

3.2.3 Searching for regression model parameters

So how do we find the parameters θ ? The most logical would be to choose the one maximizing the probability of seeing the sample, also known as the maximum likelihood estimate (MLE), they might not be correct because we only have a random sample of the population, not the whole population of pairs, but its best we can do with the sample we have

$$\hat{\theta} = \operatorname{argmax}_{\theta} p(\{y_i\}_{i=1}^N | \{x_i\}_{i=1}^N; \theta). \quad (3.22)$$

This is a very "closed" form expression, with all variables in one probability. Suppose we assume the sample pairs (y_i, x_i) are drawn independently. In that case, we can transform this to a slightly more "open" form expression, separating the probabilities for each ²

²We denote the separate probabilities p_i as they do not have to be identical, only independent.

$$p(\{y_i\}_{i=1}^N | \{x_i\}_{i=1}^N; \theta) = \prod_{i=1}^N p_i(y_i | x_i; \theta). \quad (3.23)$$

Another helpful trick is then to transform the probability using a logarithm, as the *argmax* operator is invariant to strictly rising transformations ³, which transforms the product into a sum which is much more tangible for the optimization as well as simplifying the expression for an extensive range of probability distributions

$$\operatorname{argmax}_{\theta} \prod_{i=1}^N p_i(y_i | x_i; \theta) = \operatorname{argmax}_{\theta} \sum_{i=1}^N \log(p_i(y_i | x_i; \theta)). \quad (3.24)$$

3.3 Regression models

This section will introduce the basic regression models building up to the GARCH model as one of the most important special cases motivating the GAS model.

3.3.1 Simple linear regression model

The simplest regression model for the relation $x \rightarrow y$ is the linear model parameterized by intercept β_0 and slope β_1 (24, p 22)

$$y = \beta_1 x + \beta_0 + e. \quad (3.25)$$

The goal of the regression would be to set the fitted values $\hat{y}$ as close as possible to the measured values y by "fitting" the model parameters β_0 and β_1 . Perhaps the most common metric would be the sum of squared error terms (these correspond to the disturbances).

$$e = y - \hat{y} = y - \beta_1 x - \beta_0. \quad (3.26)$$

Now we could develop an estimator just by minimizing the sum of squared error terms, but it is interesting to develop the estimator using the probability toolkit we have described. Let us assume that disturbance e is not correlated with the independent variable x , has normal distribution $\mathcal{N}$, with zero mean $\mu = 0$ and constant variance σ^2 (27).

The first assumption that e is not correlated to x , also known as exogeneity, assumes that we have no confounding variables $e \leftarrow z \rightarrow x$. The second assumption is that the distribution is normal. Since we can assume that there are many factors that we cannot measure, we can leverage the central limit theorem that says that a sum of many distributions converges to

³Strictly increasing transformation $T : X \rightarrow Y$ meaning $x < y \Leftrightarrow T(x) < T(y); \forall x, y$.

a normal one, thus justifying this assumption⁴. The third assumption is that the disturbance has zero mean $E[e] = 0$. We can justify this assumption because the intercept variable β_0 is included in the model and should cover this bias in e if we are minimizing the error. The last assumption, variance $\text{Var}[e]$ is constant, is known as homoscedasticity. This assumption is one of the harder ones to justify. However, assuming that the measurement conditions do not change, the variance should not change either. This assumption is similar to exogeneity (27)⁵.

With these and the assumption of independently drawn samples, we can develop the estimator, also known as the ordinary least squares estimator (OLS)

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{\beta_0, \beta_1}{\operatorname{argmax}} \prod_{i=1}^N p(y_i | x_i; \beta_0, \beta_1) \quad (3.27)$$

$$= \underset{\beta_0, \beta_1}{\operatorname{argmax}} \prod_{i=1}^N p(y_i | x_i; \beta_0, \beta_1) \quad (3.28)$$

$$= \underset{\beta_0, \beta_1}{\operatorname{argmax}} \prod_{i=1}^N \mathcal{N}(y_i - \beta_1 x_i - \beta_0, \sigma^2) \quad (3.29)$$

$$= \underset{\beta_0, \beta_1}{\operatorname{argmax}} \prod_{i=1}^N \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_1 x_i - \beta_0)^2}{2\sigma^2}\right) \quad (3.30)$$

$$= \underset{\beta_0, \beta_1}{\operatorname{argmax}} \sum_{i=1}^N \log\left[\frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_1 x_i - \beta_0)^2}{2\sigma^2}\right)\right] \quad (3.31)$$

$$= \underset{\beta_0, \beta_1}{\operatorname{argmax}} \sum_{i=1}^N -\frac{(y_i - \beta_1 x_i - \beta_0)^2}{2\sigma^2} - \log(\sqrt{2\pi}\sigma) \quad (3.32)$$

$$= \underset{\beta_0, \beta_1}{\operatorname{argmin}} \sum_{i=1}^N (y_i - \beta_1 x_i - \beta_0)^2. \quad (3.33)$$

As we can see from the last expression, we do not even need to know the value of variance σ^2 (even though we could use the MLE to estimate it). We are just left with the sum of squared errors. To get a close solution of the parameters, we can use mathematical optimization

$$L = \sum_{i=1}^N (y_i - \beta_1 x_i - \beta_0)^2, \quad (3.34)$$

$$\frac{\partial L}{\partial \beta_0} = -2 \sum_{i=1}^N (y_i - \beta_1 x_i - \beta_0) = 0, \quad (3.35)$$

$$\frac{\partial L}{\partial \beta_1} = -2 \sum_{i=1}^N x_i (y_i - \beta_1 x_i - \beta_0) = 0. \quad (3.36)$$

⁴Furthermore, this normality assumption is not even necessary for the consistency and unbiasedness of the estimator, it is only useful to develop the model and allow for statistical hypothesis testing and confidence intervals (27).

⁵For further information regarding the OLS assumptions, reference chapter 2 in Woolridge (24).

First, we extract β_0 by removing -2 and rearrange the expression using $N\bar{y} = \sum_{i=1}^N y_i$ and $N\bar{x} = \sum_{i=1}^N x_i$ (28)

$$N\beta_0 = N\bar{y} - N\beta_1\bar{x}, \quad (3.37)$$

$$\beta_0 = \bar{y} - \beta_1\bar{x}. \quad (3.38)$$

Second, we substitute β_0 and expand the sum from the second partial derivation (28)

$$0 = \sum_{i=1}^N x_i y_i - (\bar{y} - \beta_1 \bar{x}) x_i - \beta_1 x_i^2, \quad (3.39)$$

$$0 = \sum_{i=1}^N x_i y_i - \bar{y} \sum_{i=1}^N x_i + \beta_1 \bar{x} \sum_{i=1}^N x_i - \beta_1 \sum_{i=1}^N x_i^2. \quad (3.40)$$

Now we just extract the formula for β_1 , and we are done (28)

$$\beta_1 = \frac{\sum_{i=1}^N x_i y_i - N\bar{x}\bar{y}}{\sum_{i=1}^N x_i^2 - N\bar{x}^2} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}. \quad (3.41)$$

For this simple model, we were able to get a closed solution for the estimation of the parameters from data. A closed solution might not be available for more complicated models, and a suitable numerical optimization method has to be used for estimating the parameters from the MLE objective.

3.3.2 Autoregressive model

Regression does not have to be only applied to model static data. It can also be used to model dynamic variables; these data points are also known as time series. With this concept, we are trying to model y_t using information from previous $y_{t-1}, y_{t-2}, \dots$ samples (only previous because we are assuming causal dynamics of the variable, future cannot influence the past).

One such model is an autoregressive model (AR(p)) (29) of order p defined as

$$y_t = \sum_{i=1}^p \theta_i y_{t-i} + \epsilon_t, \quad (3.42)$$

where disturbance ϵ_t assumed to be a white noise process (25).

This model is straightforward, but the idea behind it is powerful. The current value of the variable is influenced by its previous values. This means that any change (shock) now will impact the values infinitely far into the future (25).

3.3.3 Moving average model

Another model used for time series analysis is the moving average model (MA(q)) (29) of order q (25)

$$y_t = \mu + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t. \quad (3.43)$$

This model has a different idea to the AR model, the previous values of Y_t do not influence the value of the variable Y_t , but the previous measurements do influenced it and as such previous disturbances ϵ_{t-i} as well as the current disturbance ϵ_t . This means that change (shock) in value will only temporally change the values for q steps.

3.3.4 Autoregressive moving average model

Autoregressive moving average (ARMA) (30) of order p and q is a combination of the AR and MA models, with the definition

$$y_t = \sum_{i=1}^p \varphi_i y_{t-i} + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t. \quad (3.44)$$

This model can be, for example, used to model stock prices, as internal dynamics and external shocks can influence these.

There is an extension for this model called ARMAX(p, q, b) (25) (Autoregressive–moving–average model with exogenous inputs) model, which allows for exogenous (external) variables x_t to influence the value of the variable, thus allowing additional information to be used to model the variable

$$y_t = \sum_{i=1}^p \varphi_i y_{t-i} + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \sum_{i=0}^b \eta_i x_{t-i} + \epsilon_t. \quad (3.45)$$

3.3.5 Autoregressive conditional heteroskedasticity

The autoregressive conditional heteroskedasticity (ARCH(q)) model (7) introduces an important idea to the time series analysis. The parameters of the disturbance distributions are not static. They change over time with their own dynamic, which is based on the previous

values. We can define the process as follows

$$y_t = \sigma_t \epsilon_t, \quad (3.46)$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i y_{t-i}^2, \quad (3.47)$$

$$\epsilon_t \sim (\mu = 0, \sigma^2 = 1), \quad (3.48)$$

where the error term ϵ_t can be any distribution with zero mean and variance 1. As such the conditional mean for ARCH(1) of y_t is also 0

$$E[y_t | y_{t-1}, \dots] = E[\sigma_t \epsilon_t | y_{t-1}, \dots] = \sigma_t E[\epsilon_t | y_{t-1}, \dots] = 0, \quad (3.49)$$

and using the law of iterated expectations (LIE) (31) $E[y_t] = E[E[y_t | y_{t-1}, \dots]] = 0$ the unconditional mean is also 0, so the ARCH(1) process has zero mean (32). Using LIE we can also get that the ARCH(1) process is serially uncorrelated

$$E[y_t y_{t-1}] = E[E[y_t y_{t-1} | y_{t-1}, \dots]] = E[y_{t-1} E[y_t | y_{t-1}, \dots]] = 0, \quad (3.50)$$

and $\text{Cov}[y_t, y_{t-1}] = E[y_t y_{t-1}] - E[y_t]E[y_{t-1}] = 0$, this is motivated by the efficient market hypothesis (33), as the ARCH model is used to model market volatility. (32).

3.3.6 Generalized autoregressive conditional heteroskedasticity

The generalized autoregressive conditional heteroskedasticity (GARCH) (3) is an extension of the ARCH model as a way to allow for not only the measured previous disturbances ϵ_{t-1} to influence the variance but also the previous values of the variance σ_{t-1}^2 to influence the current variance σ_t^2 . The originally defined GARCH(p, q) process has the definition

$$y_t = \beta x_t + \epsilon_t, \quad (3.51)$$

$$\epsilon_t \sim \mathcal{N}(0, \sigma_t^2), \quad (3.52)$$

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{i=1}^p \varphi_i \sigma_{t-i}^2. \quad (3.53)$$

In this definition, an exogenous variable x_t was also included, and the σ_t is now based on the disturbance ϵ_t instead of the value y_t as it no longer has zero mean. The GARCH model is also used to model volatility in financial markets. The advantage of including previous σ_{t-1} terms allows for the variance to be less "bustly", quickly rising and dropping.

The ARCH and GARCH models have many extensions, introducing exogenous variables or allowing different disturbance distributions such as t-GARCH (10).

3.4 Generalized autoregressive score model

3.4.1 Definition

As mentioned in the introduction, the generalized autoregressive score process is defined as

$$\mathbf{y}_t \sim p(\mathbf{y}_t | \mathbf{f}_t, \mathbf{x}_t; \boldsymbol{\theta}), \quad (3.54)$$

$$\mathbf{f}_{t+1} = \boldsymbol{\omega} + \sum_{i=1}^q \boldsymbol{\alpha}_i \mathbf{s}_{t-i+1} + \sum_{i=1}^p \boldsymbol{\varphi}_i \mathbf{f}_{t-i+1}, \quad (3.55)$$

$$\mathbf{s}_t = \mathbf{S}_t \nabla_t, \nabla_t = \frac{\partial \ln p(\mathbf{y}_t | \mathbf{f}_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial \mathbf{f}_t}, \quad (3.56)$$

where q and p is the score order and autoregressive order, $\mathbf{y}_t$ is the observed variables, $p(\cdot)$ is the probability density/mass function, $\mathbf{f}_t$ are the time varying parameters of the distribution, $\mathbf{x}_t$ exogenous variables, $\boldsymbol{\theta}$ is vector of static parameters of the model.

Furthermore, $\boldsymbol{\omega}$ is a vector of constants, $\boldsymbol{\alpha}_i$ for $i \in \{1, \dots, q\}$ are matrices of score coefficients, $\boldsymbol{\varphi}_i$ for $i \in \{1, \dots, p\}$ are matrices of autoregressive coefficients. All three of $\boldsymbol{\omega} = \boldsymbol{\omega}(\boldsymbol{\theta})$, $\boldsymbol{\alpha}_i = \boldsymbol{\alpha}_i(\boldsymbol{\theta})$, $\boldsymbol{\varphi}_i = \boldsymbol{\varphi}_i(\boldsymbol{\theta})$ are based on a vector of static parameters of the model $\boldsymbol{\theta}$.

The function of the past data (score) $s_t = s(\mathbf{f}_t, \mathbf{y}_t, \mathbf{x}_t, \boldsymbol{\theta})$ is based on the time-varying parameters $\mathbf{f}_t$, observations $\mathbf{y}_t$, exogenous variables $\mathbf{x}_t$ and the static parameters $\boldsymbol{\theta}$.

The most common choices for the scaling functions are identity $\mathbf{S}_t = \mathbf{I}$, the (pseudo)inverse of the information matrix $\mathbf{S}_t = \mathcal{I}_t^{-1}$ or square root of the (pseudo)inverse of the information of matrix $\mathbf{S}_t = \mathcal{I}_t^{-1/2}$, such that $\mathbf{S}_t^T \mathbf{S}_t = \mathcal{I}_t^{-1}$, this is usually done by Cholesky decomposition (34). The information matrix can also be interpreted as the variance of the unscaled score ∇_t

$$\mathcal{I}_t = \mathbb{E} \left[\nabla_t \nabla_t^T \right], \quad (3.57)$$

with $\mathbb{E}[\cdot]$ being the expected value, with respect to $p(\mathbf{y}_t | \mathbf{f}_t, \mathbf{x}_t; \boldsymbol{\theta})$ (2). The inverse of the information matrix $\mathbf{S}_t = \mathcal{I}_t^{-1}$ allows us to show that specific cases of GAS models are equivalent to some previously defined models. The square root of the (pseudo)inverse of the information of matrix $\mathbf{S}_t = \mathcal{I}_t^{-1/2}$ brings in some valuable statistical properties such that the score $\mathbf{s}_t$ has a unit variance.

⁶The variance of the unscaled score ∇_t has the same form as the Fisher information matrix introduced in "On the mathematical foundations of theoretical statistics" by R. A. Fisher in 1992, with respect to the variable parameters $\mathbf{f}_t$ with the probability distribution $p(\mathbf{y}_t | \mathbf{f}_t, \mathbf{x}_t; \boldsymbol{\theta})$.

3.4.2 Properties

When using an estimator, it is important to discuss its properties. One of the main properties of any estimator is consistency, the property that as the number of data points n increases, the resulting sequence of estimates $\hat{\theta}_n$ converges in probability to the true value θ_0 (35)

$$\lim_{n \rightarrow \infty} P[|\hat{\theta}_n - \theta_0| > \varepsilon] \rightarrow 0, \forall \varepsilon > 0, \quad (3.58)$$

Another useful property of an estimator is asymptotic normality. If the consistency property assures the validity of the estimator, that we eventually get the right values of θ_0 , this property assures the convergence rate is fast enough. This property can be defined as

$$\lim_{n \rightarrow \infty} \sqrt{n}(\hat{\theta}_n - \theta_0) \rightarrow^d \mathcal{N}(0, \sigma_{\theta_0}^2), \quad (3.59)$$

For the time-varying observation-driven models, there is another essential property called invertibility(36), it assures that the error estimates of the time-varying parameters $\hat{f}_t$ converge to the real values of time-varying parameters f_t (37).

The article "Maximum Likelihood Estimation for Generalized Autoregressive Score Models" (38) by F. Blasques, S. J. Koopman, and A. Lucas provide low-level conditions that allow them to show the invertibility of the score filter. With these results, the consistency and asymptotic normality of the estimator is then discussed for well-specified and misspecified GAS models.

3.4.3 Special cases

As previously mentioned, GAS model is a generalization of previously defined models, and several of the following special cases are shown.

GARCH(1,1) By choosing time-varying parameter as the conditional variance $f_t = \sigma^2$, the probability density function as $p(y_t) = \mathcal{N}(y_t; 0, \sigma^2)$ (we assume zero mean for simplicity) and appropriate scaling function $S(\sigma^2) = \mathcal{I}_t = 2(\sigma^2)^2$, we get model

$$\sigma_{t+1}^2 = \omega + \alpha(y_t^2 - \sigma_t^2) + \varphi\sigma_t^2, \quad (3.60)$$

which is equivalent to the GARCH(1,1) (3) model with the difference in the parametrization of $\varphi \leftarrow \varphi - \alpha$ (2).

Note: If we changed p to Student t-distribution, the model would *not* be equivalent to the t-GARCH model (10). This is because of the nature of the score using a partial derivative, and this is an advantage of the GAS framework as for the fat-tailed distribution, the volatility would not change so dramatically in response to large values of $|y_t|$ (39).

ACD(1,1) By choosing the time-varying parameter as the mean duration $f_t = \mu_t = \frac{1}{\lambda_t}$, the probability density function as $p(y_t) = \text{Exp}(y_t; \frac{1}{\mu_t})$ and $S(\mu) = \mathcal{I}_t$

$$\mu_{t+1} = \omega + \alpha(y_t - \mu_t) + \varphi\mu_t, \quad (3.61)$$

which is equivalent to the ACD(1,1) (4) model with the difference in parametrization of $\varphi \leftarrow \varphi - \alpha$ (2).

ACI(1,1) By choosing the time-varying parameter as the logarithmic intensity $f_t = \log(\lambda_t)$, the probability density function as $p(y_t) = \text{Exp}(y_t; \lambda_t) = \text{Exp}(y_t; \exp(f_t))$ and $S(\mu) = \mathcal{I}_t$ and $S(f_t) = \mathcal{I}_t$ we obtain the equation

$$f_{t+1} = \omega + \alpha(1 - y \exp(f_t)) + \varphi f_t, \quad (3.62)$$

which is equivalent to the ACI(1,1) model (5).

3.4.4 Adapting to parametrization

As shown in the original paper introducing GAS model (2), GAS adapts naturally to different parametrizations of probability distributions. This property becomes useful; for example, when enforcing the positivity of some parameters using logarithmic transformation or when working with aggregate data such as duration vs. intensity, shown in the examples of ACD and ACI models.

4. Methodology

In this chapter, we will describe the methodology of the practical part of the bachelor thesis, the simulation study. We will review the selected distributions, model parameters, provided information, and the technical details of the simulation and estimation, such as the library used and scale of the simulation (data length and repetitions). Lastly, the metric used for evaluating the precision of the estimator.

4.1 Selected models

We have chosen 11 models, all of the order GAS(1,1), with six different distributions ¹. For the 2-parameter distributions, we assume one of the distribution parameters as static (not changing in time) ². However, we do differentiate the models where the parameter is assumed to be known and when it should be estimated alongside the GAS parameters (ω , α and φ) for the time-varying distribution parameter. All time-varying parameters are in logarithmic transformation as they all have to be positive and using log. transform assures their positivity, regardless of the values of the GAS parameters. In table 4.1 ³ you can see an overview of the selected models.

distr.	time var. param.	static param.	fixated static param.
normal	log(variance)	mean	yes
normal	log(variance)	mean	no
Student t (d.f. 5)	log(variance)	mean	yes
Student t (d.f. 5)	log(variance)	mean	no
exponential	log(scale)	-	-
gamma	log(scale)	shape	yes
gamma	log(scale)	shape	no
Weibull	log(scale)	shape	yes
Weibull	log(scale)	shape	no
neg. binomial	log(mean)	dispersion	yes
neg. binomial	log(mean)	dispersion	no

Table 4.1: Table of selected models for simulation

¹Inspired by "Predicting Time-Varying Parameters with Parameter-Driven and Observation-Driven Models" (18) by S. J. Koopman, A. Lucas.

²The static assumption of one of the parameters is necessary for the computation speed as we are estimating a large number of models.

³We have chosen the Student t-distribution to represent distributions with heavy tails. Because of this, we select the degrees of freedom parameter as 5, which is relatively small, and as such, the t-distribution should still show different properties compared to the normal distribution.

4.2 Model parameters

For the selection of model parameters, we should choose values that sufficiently represent the range of possibilities of the GAS model so as not to introduce bias into the simulation results. For the simulation length, we have chosen values (100, 250, 500, 1000, 2000) ⁴, an almost logarithmic scale doubling in each step. For the GAS parameters, constant ω , score coefficient α , and autoregressive coefficient φ , we chose set {0.2, 0.5, 0.8} as possible choices for all three parameters, as they are suitable to representatives for the common range of values [0, 1]. As the scaling function, the identity has been chosen for its simplicity. The following table 4.2 shows the summary of parameter values, including the static parameter values for the 2-parameter distributions ⁵.

parameter	symbol	value(s)
simulation length	N	{100, 250, 500, 1000, 2000}
constant	ω	{0.2, 0.5, 0.8}
score coefficient	α	{0.2, 0.5, 0.8}
autoreg. coefficient	φ	{0.2, 0.5, 0.8}
mean (norm, t)	μ	0
shape (gamma, Weibull)	k	2
dispersion (negbin)	$1/k$	0.25

Table 4.2: Model parameters values

With all combinations of N , ω , α , and φ , each model has 135 possible settings to cover some of the possible bias that would be introduced using just one set of parameters.

4.3 Simulation and estimation

The practical part of this thesis has been done using the *R* language (40), which is a scripting language for numerical calculations, an increasingly common choice in the field of econometrics as there are many well-documented and supported packages for statistics and regression. Specifically, a custom script has been created to set up the model parameter combinations and then run the simulation and estimation of the models, saving the estimated parameters. The GAS process simulation and the GAS model estimation are done using *gasmodel* package.

For each of the 11 models and 135 parameter combinations, we do 20 repetitions, totaling 29700 estimated models, which in total took approximately 83.2 hours of computation time

⁴Inspired by "Predicting Time-Varying Parameters with Parameter-Driven and Observation-Driven Models" (18) by S. J. Koopman, A. Lucas.

⁵For normal and t-distributions, zero mean is motivated similarly to the ARCH and GARCH model, the shape and the dispersion for gamma and Weibull and negative binary distributions, respectively, was selected to gain heavier tailed distributions comparatively to exponential and Poisson distributions.

on an i7-9700 CPU at 4.5GHz. This will be the dataset used to test the properties of the GAS models. The following dataset has 15 variables (described in table 4.3) and 29700 entries.

Note To limit the time necessary for computation, the optimization parameters were set to relative tolerance for variables 10^{-5} and a maximum of 10^5 evaluations of the loss function.

variable	description
t_sim	length of the time series in simulation and estimation
distr	probability distribution (norm, t, exp, gamma, weibull, negbin)
par	what is the time-varying parameter (var, scale, mean)
scaling	what scaling function is used, currently only unit (meaning identity)
stat_par1	value of the first static parameter, if necessary
stat_par2	value of the second static parameter, deg. of freedom in t
stat_par_fix	denoting if the static parameter is fixated in the model
true_omega	true value of the constant parameter ω
true_score	true value of the score coefficient α
true_autoreg	true value of the autoregression coefficient φ
est_omega	estimate of the constant parameter ω
est_score	estimate of the score coefficient α
est_autoreg	estimate of the autoregression coefficient φ
est_statpar	estimate of the static parameter, only used when it is not fixated
execution_time	time of the execution of simulation and estimation in seconds

Table 4.3: Description of dataset variables

4.3.1 Package *gasmodel*

The *gasmodel* (ver. 0.2.0) package by V. Holý (41), is an R language package being developed at the Department of Econometrics at FIS VŠE, used for estimation, simulation, and forecasting GAS models. Specifically, the functions *gas_simulate*, and *gas* functions have been used in this thesis. This package is relatively easy to work with, allowing for the full generality of the GAS model framework to be used.

In the original outline of this thesis, the Poisson distribution was also included in the list of distributions. However, often, the simulated data was invalid for this distribution, N/A values were returned for some combination of parameters, possibly due to a bug or incorrect use of the function.

4.4 Model performance measure

A simple mean square error MSE is used to measure the model performance. It is the mean of squared errors of parameter estimates $\hat{\theta}$ from the real values θ . If the estimator was unbiased $E[\theta - \hat{\theta}] = 0$, it could also be interpreted as the estimator's variance

$$MSE = \frac{1}{N} \sum_{i=1}^N e_{\theta}^2 = \frac{1}{N} \sum_{i=1}^N (\theta - \hat{\theta})^2 \quad (4.1)$$

5. Results

This chapter presents the results of the practical part of the thesis, the analysis of the simulated data. Multiple comparisons will be presented for different crosssections of the data generated in this simulation study.

5.1 PDF estimate for errors

Firstly we create a PDF estimate for the errors of model parameters and present it using graphs in figures 5.1, 5.2, 5.3, 5.4 and 5.5. This graph shows the estimator's behavior across distributions and simulation lengths. This is a good baseline before doing more analysis and aggregating data.

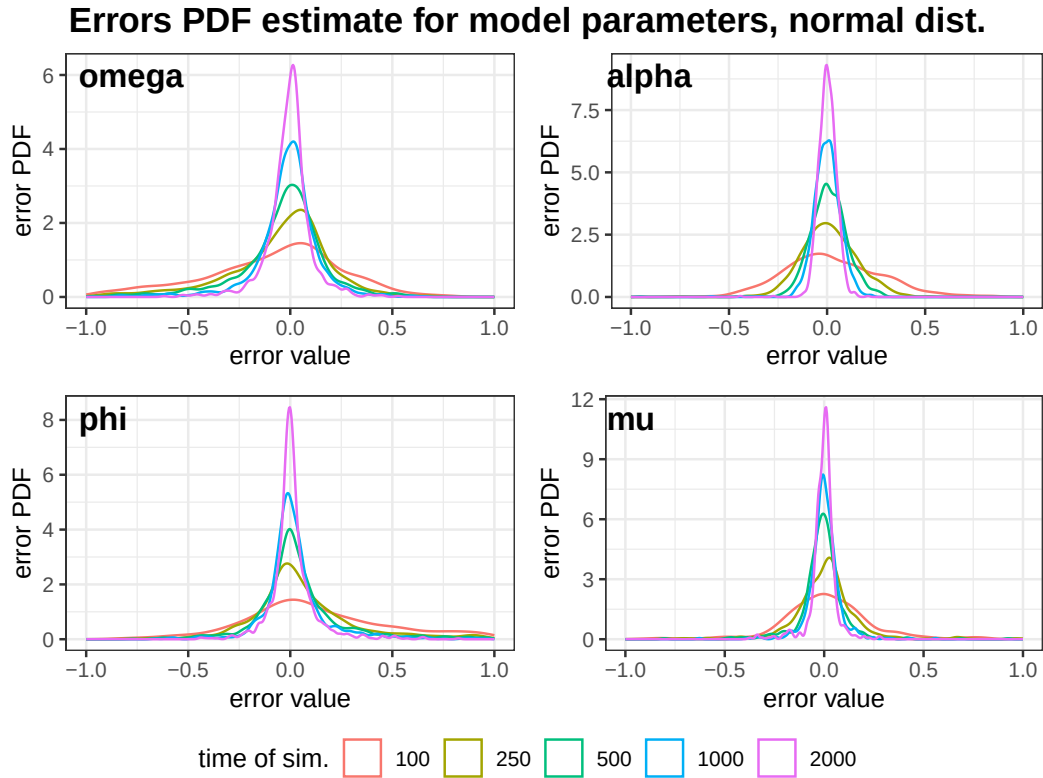


Figure 5.1: Error PDF estimate for model parameters, normal distribution

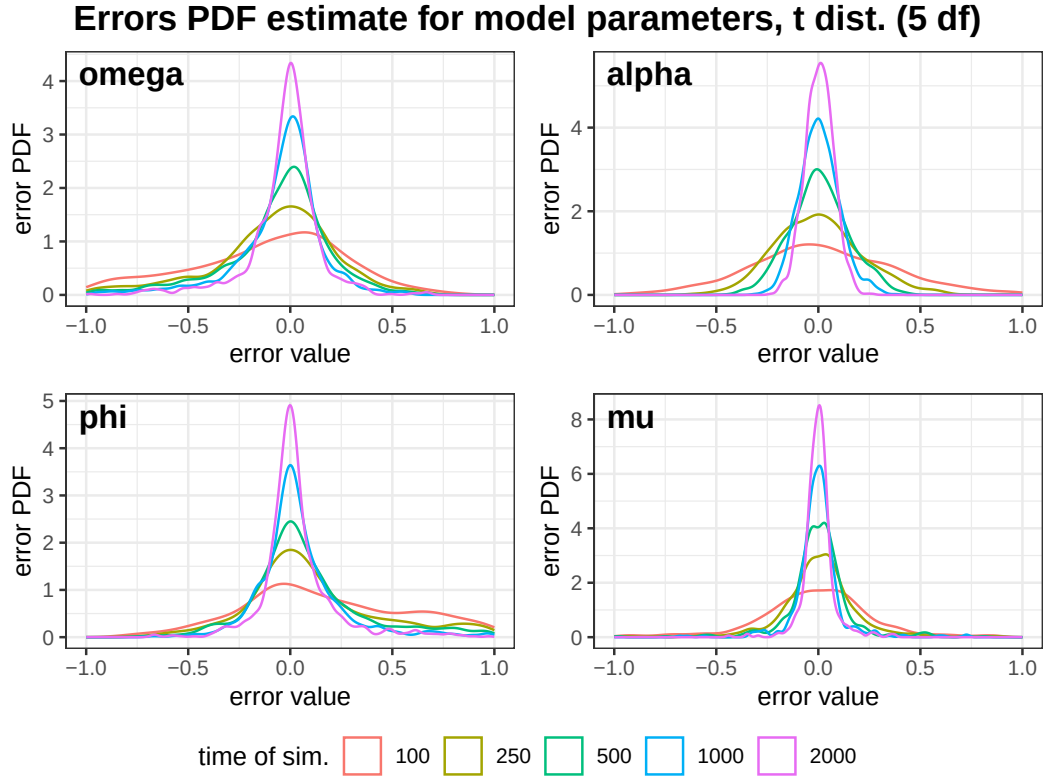


Figure 5.2: Error PDF estimate for model parameters, t distribution (5 d.f.)

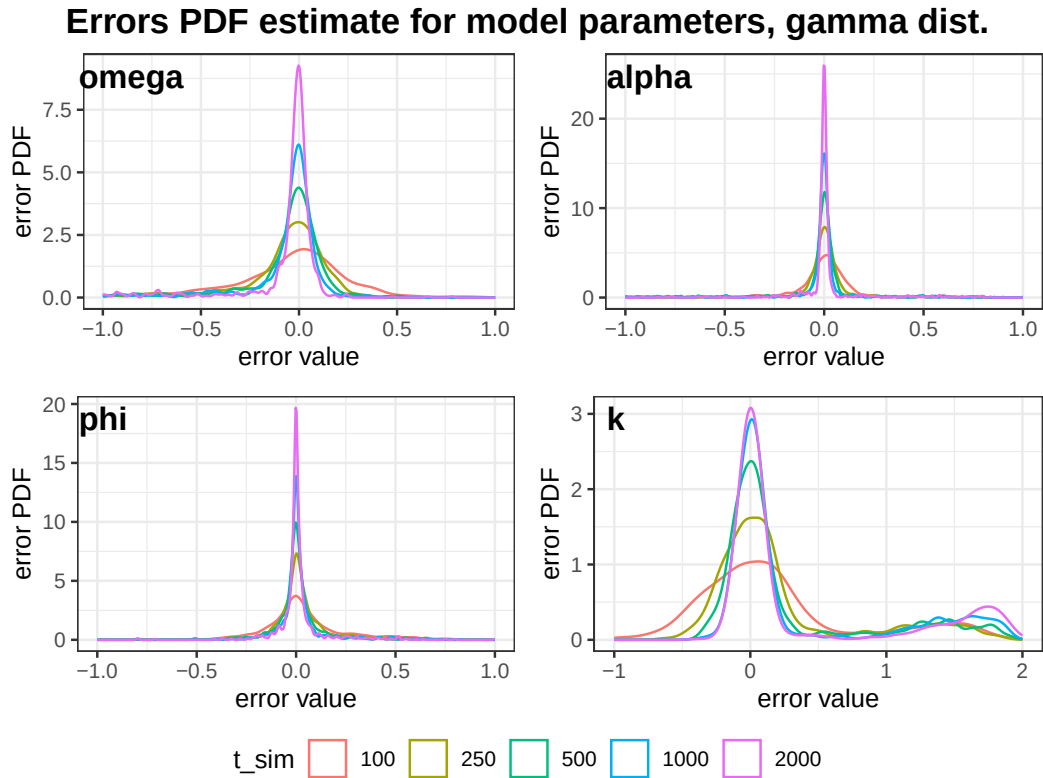


Figure 5.3: Error PDF estimate for model parameters, gamma distribution

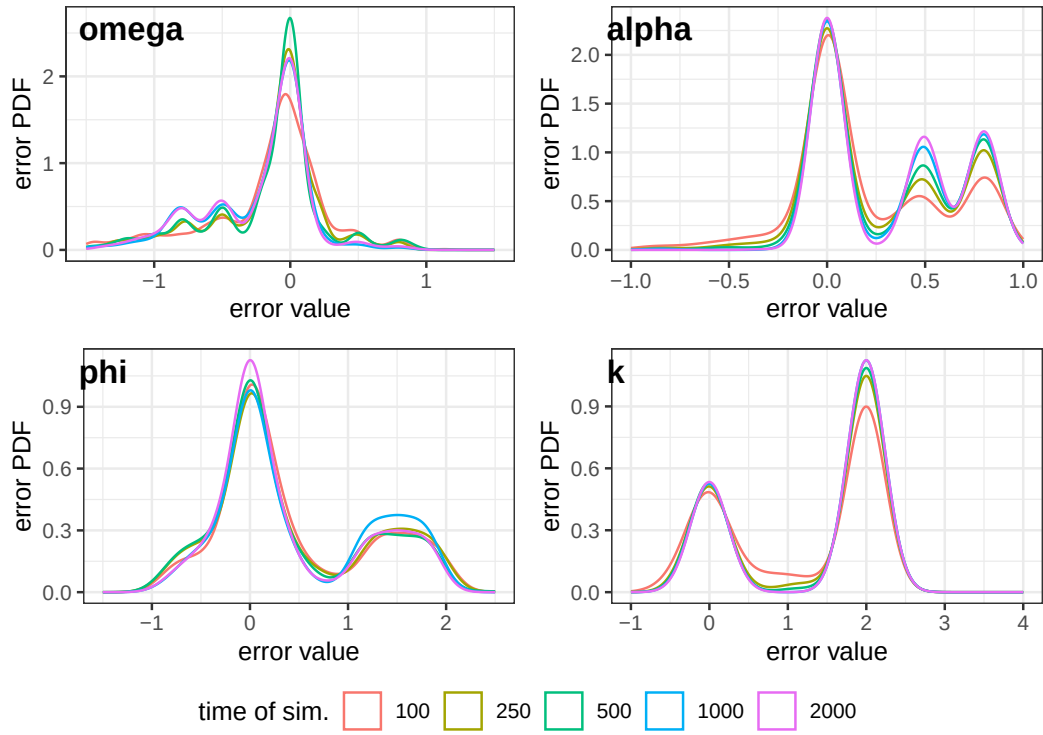
Errors PDF estimate for model parameters, weibull dist.

Figure 5.4: Error PDF estimate for model parameters, Weibull distribution

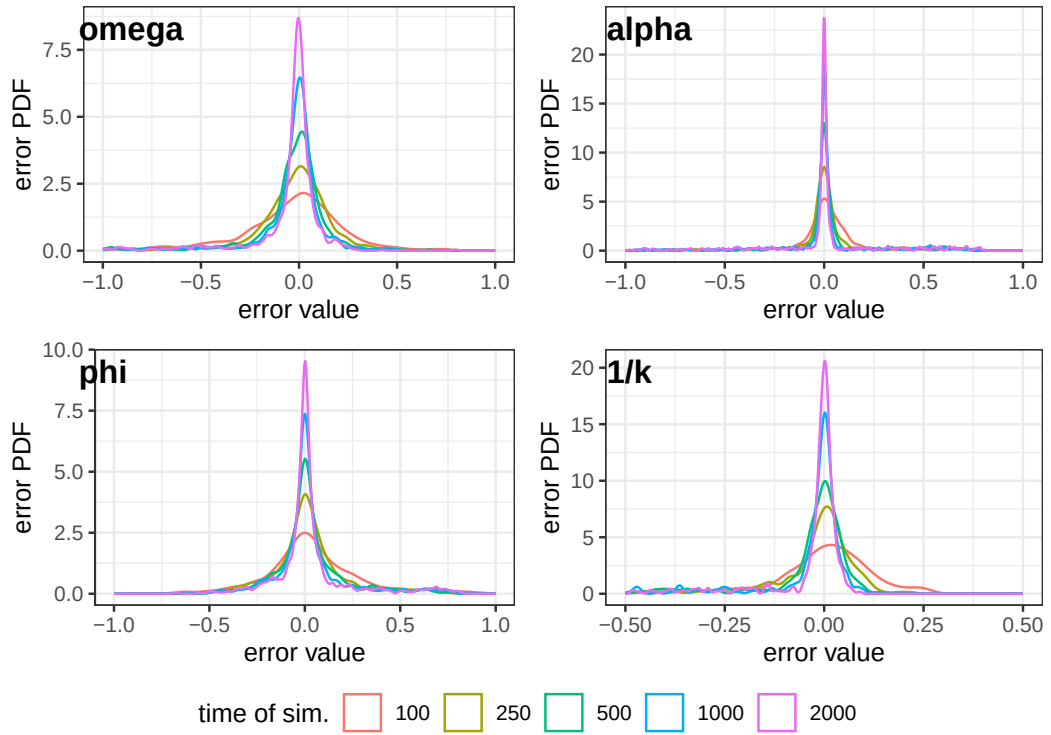
Errors PDF estimate for model parameters, neg. binomial dist.

Figure 5.5: Error PDF estimate for model parameters, negative binomial distribution

5.2 Influence of fixated parameters and data length on MSE

Secondly, the information on how the fixating of the static parameter and the data length influences MSE is presented in multigraph 5.6. For the exponential, only the data length is shown as it is a 1-parameter distribution. To reduce the bias of outliers, for example, due to insufficient computation time, only 5% to 95% quantile values of the estimation errors are used to calculate MSE. This graph is to show the consistency of the estimator and to investigate its properties.

5.3 Influence of score coefficient on MSE for gamma and neg. binomial distribution

To more closely investigate the behavior of the estimator for gamma and negative binomial distribution, the graph in figure 5.7 shows the relationship between the MSE and the value of the score coefficient and data length. Again, only 5% to 95% quantile values of the estimation errors are used to calculate MSE.

5.4 Model parameter estimate PDF for Weibull distribution

To more closely investigate the estimator for Weibull distribution, the PDF for the model parameter estimates is presented in graph figure 5.8. Alongside the graph, there is also presented the out-of-bounds statistic (OoB), denoting how much percentage of data points is not presented in the graph. This shows how many extreme (outlier) values are in the distribution.

5.5 MSE for converging models

The MSE for converging models, normal, Student t, and exponential distributions across all parameters and gamma, and negative binomial distributions for $\alpha = 0.2$, is shown in graph figure 5.9. Again, to reduce the bias of outliers, only 5% to 95% quantile values of the estimation errors are used to calculate MSE. This graph is a more readable comparison of how big a sample we need to achieve a certain precision of a selected parameter. To achieve a more precise comparison between parameters, the scale and position of the y-axis is constant for all parameters.

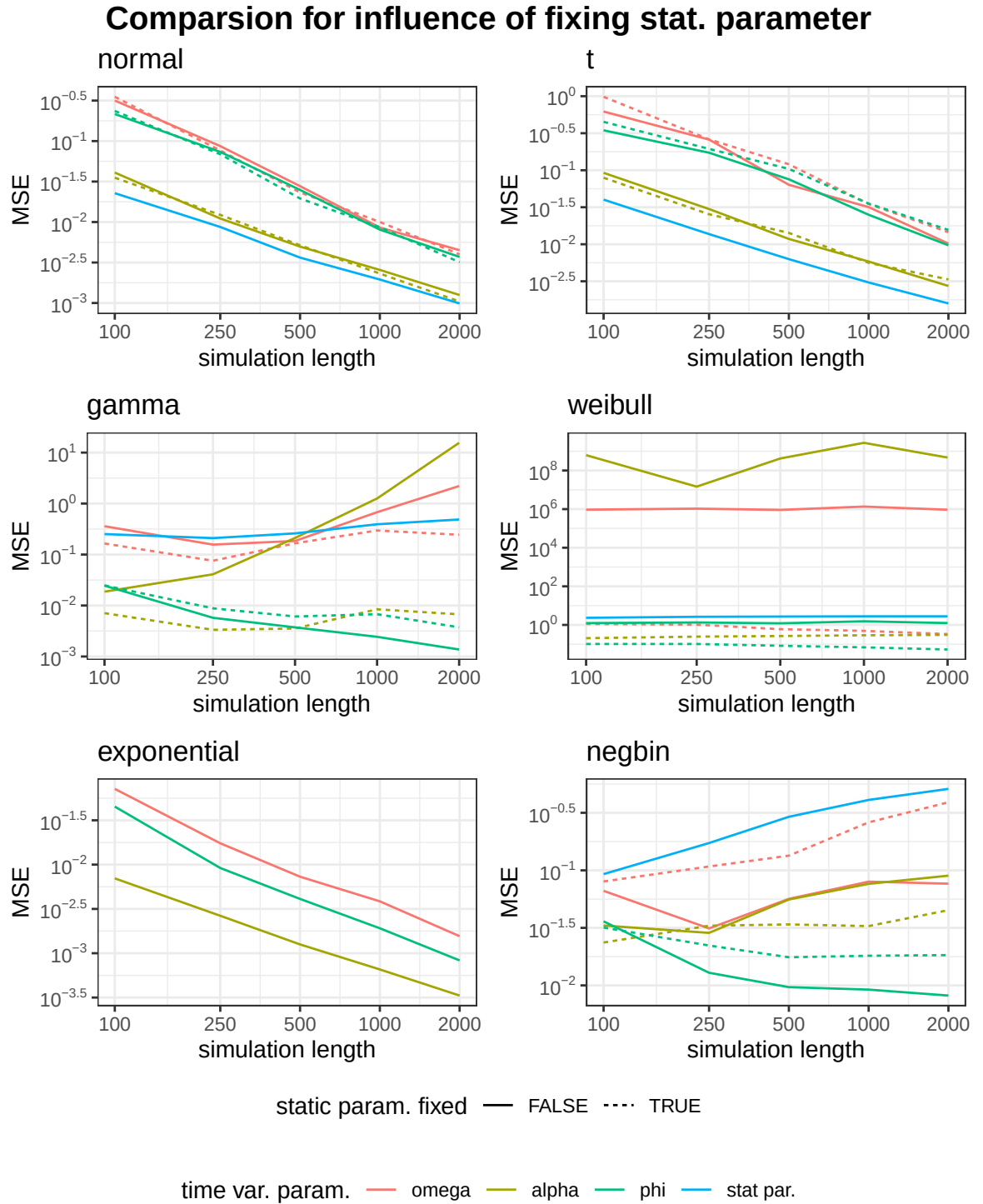


Figure 5.6: MSE with respect to the static parameter and data length

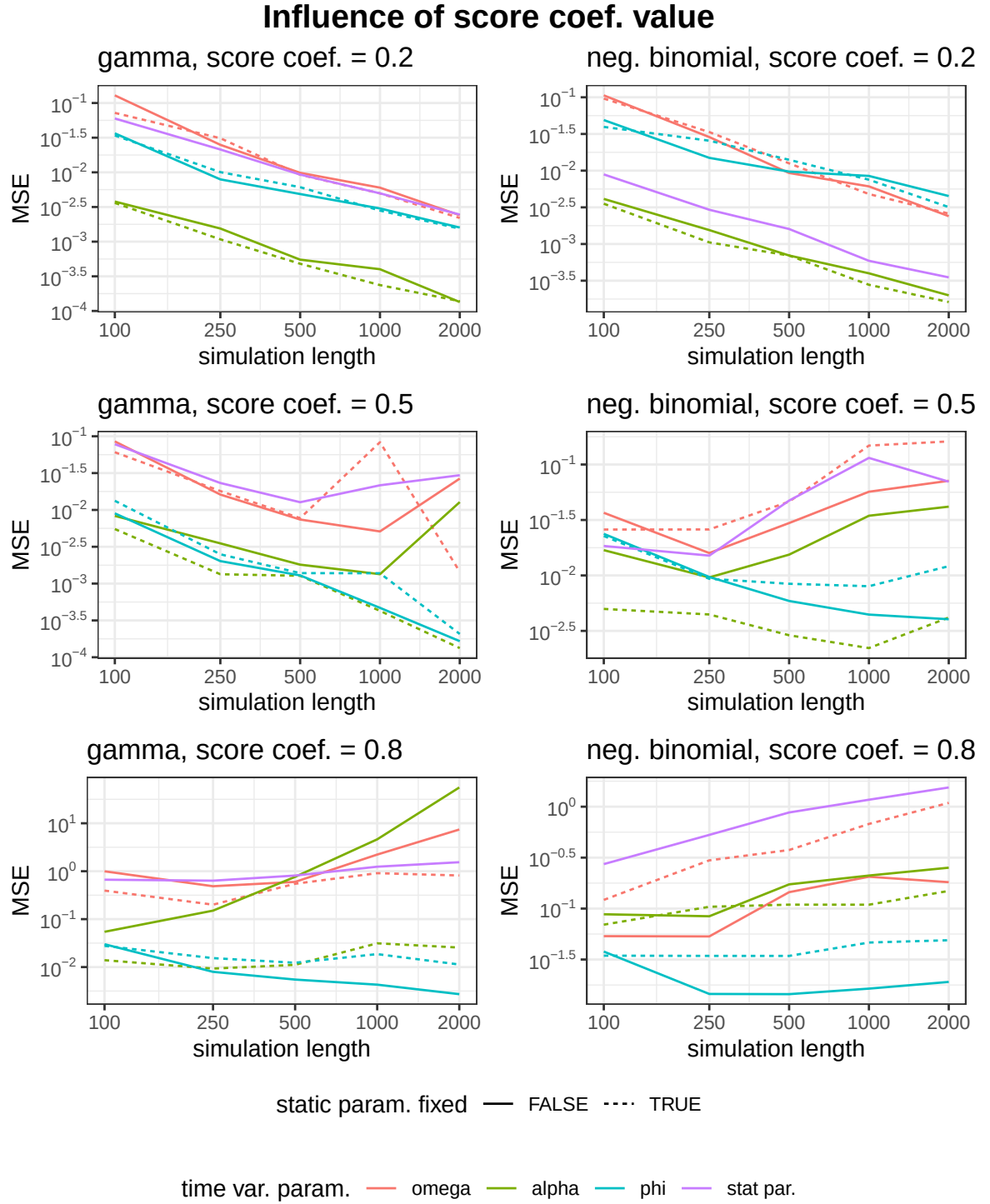


Figure 5.7: MSE with respect to the score coefficient and data length for gamma and negative binomial distribution

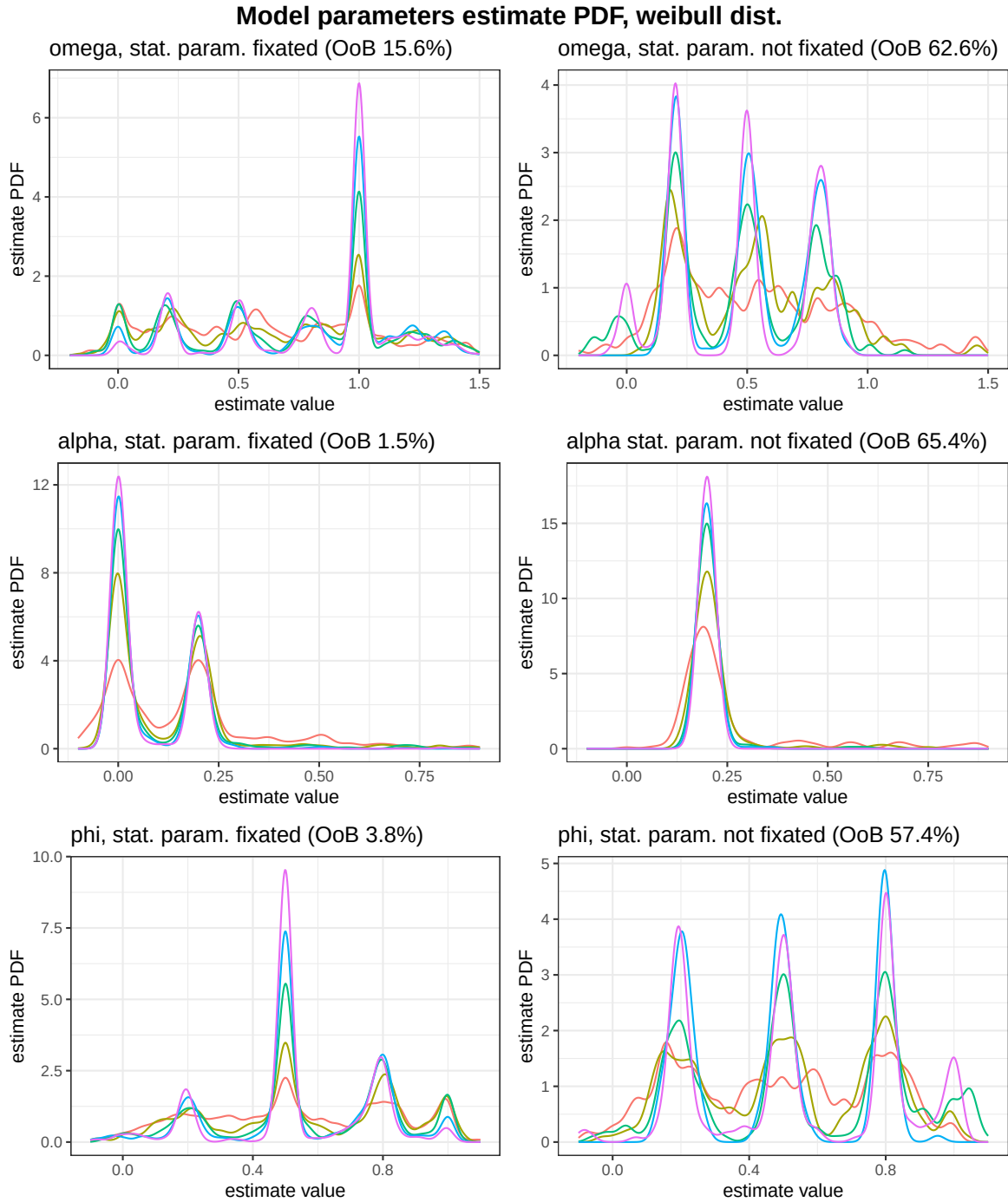


Figure 5.8: PDF of model parameter estimates for Weibull distribution

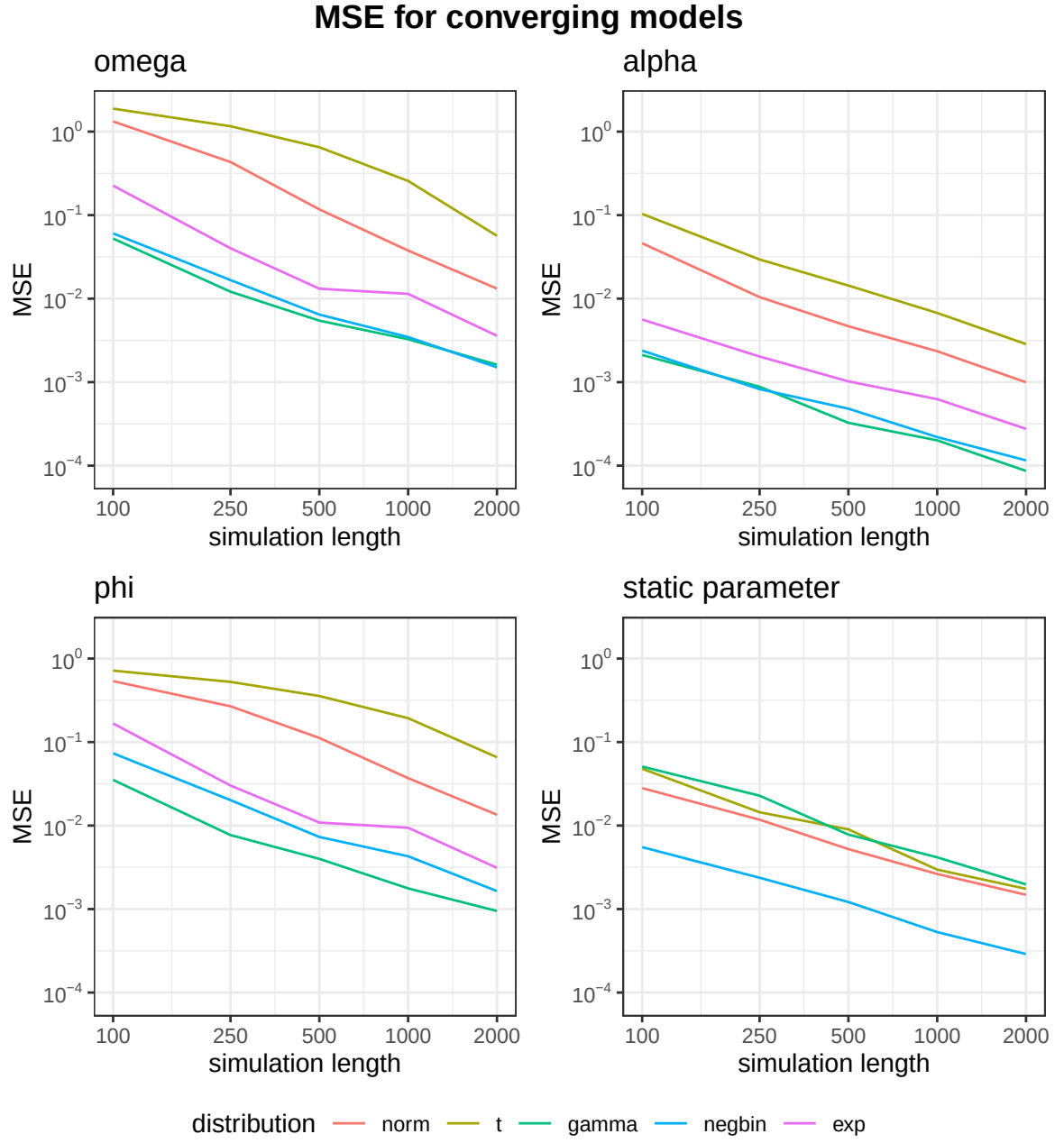


Figure 5.9: MSE for converging models by parameter with respect to distribution and data length

6. Discussion

In this chapter, we will discuss the results presented in the previous chapter. We will review the expected results, demonstrating the consistency of some of the defined models as well as comparing the precision of their model parameter estimation, outlining the unexpected results for some distributions, and hypothesize why this deviation from expectations is present.

6.1 Estimator properties

As seen from PDF graphs in figures 5.1, 5.2, 5.3, 5.4 and 5.5 and the MSE graphs in figure 5.6, the models for normal, Student t, and exponential distribution are behaving as expected, with the greater length of simulation (number of samples) the estimated values converge to the true values (errors converge to zero). When investigating the MSE graph for estimator consistency, it is important to look at the trends; the absolute values of MSE are not as relevant.

From PDF graphs in figures 5.3 and 5.5, it would seem that the models for gamma and negative binomial distribution converge as well. However, upon investigating the graphs in figure 5.6 the MSE for these distributions diverges, their MSE increases with the number of samples. The PDFs in figure 5.3 and 5.5 look proper because the outliers are out-of-bounds of the graph.

Even more extreme is the model for Weibull distribution. We can see right from the PDF in figure 5.4 that the estimator is not consistent. However, upon investigating the MSE graph in figure 5.6 for the Weibull distribution, one can see the MSE does not diverge; it stays mostly constant. This would suggest that this is due to a different effect compared to the gamma and negative binomial distributions.

6.2 Divergence of MSE of gamma and negative binomial models

To investigate the effect of divergence of gamma and neg. binomial models, the MSE graph in figure 5.7 shows the influence of score coefficient values α on the estimator. As one can see, for the low value of $\alpha = 0.2$, both estimators behave properly, and the MSE converges to 0, similar to the normal, Student t, and exponential distribution. However, for the values $\alpha = 0.5$ and $\alpha = 0.8$, the MSE diverges.

One hypothesis for this effect is that values 0.5 and 0.8 for the score coefficient α might violate the conditions for the consistency of the model or invertibility of the score filter¹ and

¹"Maximum Likelihood Estimation for Generalized Autoregressive Score Models" (38) by F. Blasques, S. J. Koopman, and A. Lucas discuss the consistency and other properties of the GAS models in detail.

thus the models might not converge to the correct values.

6.3 Stagnation of MSE for Weibull distribution model

As seen from the graph in figure 5.6, we can see that the MSE stagnates on significantly higher values than the other distributions. The reason is demonstrated in the PDF for estimates graphs for Weibull distribution in graphs figure 5.8. As one can see, for the fixated model, the parameters stay relatively within bounds and have the expected peaks at 0.2, 0.5, and 0.8. However, they very often converge to incorrect values of $\omega = 1$ and $\alpha = 0$. For the free (non-fixated) model, more than half of the estimates are out of bounds. We cannot analyze the properties of the model as most of the estimates diverge to extreme values.

The most probable explanation for this is an issues connected to the numerical optimization of the model objective. It is a complicated nonlinear function whose optimization greatly depends on the initial values, and thus converging to the theoretical model value is not guaranteed. Within limited computation time, the optimization algorithm might not find the right solution.

6.4 Static parameter fixating

Another question to discuss is if fixating the static parameter, thus providing additional information to the model, improves the estimator's performance. An overview of the static parameters is shown in table 4.2. As seen from the MSE graph for the converging models for normal and Student t distribution as 5.6 well as the MSE graph for gamma and negative binomial distributions, when converging with $\alpha = 0.2$ in figure 5.7, fixating the static parameter does not significantly improve the MSE, often the free (non-fixated) model performs even better than the fixated one.

The only real impact where fixating the static parameter improved MSE was the Weibull distribution, as seen in the graph in figure 5.6, as well as reducing the number of OoB estimates, as seen in the graph in figure 5.8, however as this model did not converge, it is not proper to conclude information from the Weibull distribution model.

6.5 Precision of the estimator

When we compare the precision of the estimators, figure 5.9 presents the graph of MSE for the converging models, normal, Student t, and exponential distributions for all parameter combinations and gamma and negative binomial distributions for $\alpha = 0.2$, across sample size and respectively for all parameters, constant ω , score coefficient α , autoregressive coefficient φ and the static parameter (described in table 4.2).

The Student t-distribution model (with 5 degrees of freedom) requires approximately 3-times as many samples to estimate the model parameters with the same precision as the normal distribution model. This is true for all of the model parameters, with the exception of the static mean estimation, where this precision discrepancy is not as prevalent but still present. A similar conclusion can be drawn from the comparison of error PDF estimates in figures 5.1 and 5.2, where the curves for the Student t-distributions have smaller peaks and are wider.

The estimator for the intensity and duration models, exponential, gamma, and negative binomial distributions, is even more precise, compared to the variance models for normal and Student t-distributions. From the graph, it seems that the exponential distribution estimator is less precise than gamma and negative binomial estimators. However, this could be caused by the fact that the MSE metric for gamma and negative binomial distribution is only shown for $\alpha = 0.2$ as opposed to the exponential one. Because of this limitation, we cannot confidently say that the gamma and negative binomial estimators are more precise.

When we compare the precision of model parameter estimation, the score coefficient α is generally the most precise to estimate, followed by the autoregressive coefficient φ and then closely followed by the constant ω as the least precise parameter. Comparing the static parameter precision is harder to justify as the static parameter is not present in all models, and the interpretation of its values differs between distributions (explained in table 4.2). However, we can see that the rate of convergence (slope of the line) is similar to the other model parameters across distributions.

Conclusion

The two main goals of this thesis have been fulfilled. Firstly, the GAS model framework has been introduced together with the relevant sources regarding this issue, and the theoretical background for time series analysis has been presented alongside an overview of time-varying parameter models and their connection to the GAS framework, and discussing the theoretical properties of the GAS model.

Secondly, a large sample of estimated GAS models has been created using simulated data and was analyzed, empirically showing the consistency and precision of some of the presented models or further explaining the possible reasons why some the models have not converged, as well as testing the developing *gasmodel* R package by V. Holý at the Department of Econometrics FIS VŠE.

The main limitation of this work is the limited range of parameters used for the simulated data. A more extensive study would possibly also show the influence of different scaling functions as well as multivariate distributions. Another possible extension is a further analysis of factors that empirically cause the divergence of the model, thus providing a guideline to the limitation of the GAS framework.

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Appendices

A. Mathematical notation table

notation	explanation
$\mathbf{x}$	vector
$\mathbf{A}$	matrix
$\{x_k\}_{k=i}^j$	sequence $\{x_i, \dots, x_j\}$
$P[X Y]$	probability of X with condition Y
$E[X Y]$	expected value of X with condition Y
$p(x y; \theta)$	prob. mass or density function of x with condition y and parameters θ
$x \sim D(\theta)$	x is drawn from probability distribution D with parameter θ
$x \sim p(x)$	x is drawn from prob. distribution with prob. mass or density function p
$\lim x \rightarrow^d D$	x has distribution D in the limit
$\mathbb{N}_0$	non-negative integers, including zero
$\mathbb{R}_0^+$	non-negative real number, including zero

B. Source code for the simulation study (R language script)

```
library(gasmodel)

## dist options

# norm - var
# t - var

# exp - scale
# gamma - scale
# weibull - scale

# pois - mean
# negbin - mean

# prepare dataset of instances to simulate
setup_gas_sim <- function(N=1, t_sim=c(500), distr="norm", par=NA,
                          scaling="unit", stat_par1=NA, stat_par2=NA,
                          stat_par_fix = TRUE, true_omega=c(0),
                          true_score=c(0.5), true_autoreg=c(0.5))
{
  col_names = c("t_sim", "distr", "par", "scaling",
                "stat_par1", "stat_par2", "stat_par_fix",
                "true_omega", "true_score", "true_autoreg")

  if ((distr == "norm") || (distr == "t")) {
    if (any(is.na(stat_par1))) stat_par1 <- c(0)
    if (any(is.na(par))) par <- "var"
    if (distr == "t") {
      if (is.na(stat_par2)) stat_par2 <- c(log(2))
    }
  }

  else if (distr == "exp" || distr == "gamma" || distr == "weibull") {
    if (any(is.na(stat_par1))) stat_par1 <- c(1)
    if (any(is.na(par))) par <- "scale"
  }
}
```

```

else if ((distr == "pois") || (distr == "negbin")) {
  if (any(is.na(par))) par <- "mean"
  if (any(is.na(stat_par1))) stat_par1 <- c(1)
}

df <- expand.grid(t_sim, distr, par, scaling,
                 stat_par1, stat_par2, stat_par_fix,
                 true_omega, true_score, true_autoreg, 1:N)

df <- df[,1:(ncol(df)-1)]

colnames(df) <- col_names
df["est_omega"] <- NA
df["est_score"] <- NA
df["est_autoreg"] <- NA
df["est_statpar"] <- NA
df["execution_time"] <- NA

df

return(df)
}

# run the simulation on the dataset created by setup_gas_sim
run_gas_sim <- function(df)
{
  for (i in 1:nrow(df)) {
    # set parameters based on distribution

    start_time <- Sys.time()

    t_sim = df[i, "t_sim"]
    distr = as.character(df[i, "distr"])
    par = as.character(df[i, "par"])
    scaling = as.character(df[i, "scaling"])

    if (distr == "norm") {
      par_stat <- c(TRUE, FALSE)
      coef_est <- c(df[i, c("stat_par1", "true_omega",
                           "true_score", "true_autoreg")])
      if (df[i, "stat_par_fix"]) {
        coef_fix_value <- c(df[i, "stat_par1"], NA, NA, NA)

```

```

    }
    else {
      coef_fix_value <- NULL
    }
    stat_par_name <- "mean"
  }

  else if (distr == "t") {
    par_stat <- c(TRUE, FALSE, TRUE)
    coef_est <- c(df[i, c("stat_par1", "true_omega", "true_score",
                          "true_autoreg", "stat_par2")])
    coef_fix_value <- c(df[i, "stat_par1"], NA, NA, NA,
                        df[i, "stat_par2"])
    if (df[i, "stat_par_fix"]) {
      coef_fix_value <- c(df[i, "stat_par1"], NA, NA, NA,
                          df[i, "stat_par2"])
    }
    else {
      coef_fix_value <- c(NA, NA, NA, NA, df[i, "stat_par2"])
    }
    stat_par_name <- "mean"
  }

  else if ((distr == "gamma") || (distr == "weibull") ||
           (distr == "negbin")) {
    par_stat <- c(FALSE, TRUE)
    coef_est <- c(df[i, c("true_omega", "true_score",
                          "true_autoreg", "stat_par1")])
    if (df[i, "stat_par_fix"]) {
      coef_fix_value <- c(NA, NA, NA, df[i, "stat_par1"])
    }
    else {
      coef_fix_value <- NULL
    }
    if (distr == "negbin"){
      stat_par_name <- "dispersion"
    }
    else {
      stat_par_name <- "shape"
    }
  }

  else if ((distr == "exp") || (distr == "pois")) {

```

```

par_stat = NULL
coef_est = coef_est <- c(df[i, c("true_omega", "true_score",
                                "true_autoreg")])

coef_fix_value <- c(NA, NA, NA)
stat_par_name <- NULL
}

coef_est <- as.numeric(coef_est)
if (!is.null(coef_fix_value)) {
  coef_fix_value <- as.numeric(coef_fix_value)
}

sim_data <- NA
while(any(is.na(sim_data))) {
  sim <- gas_simulate(t_sim=t_sim, distr=distr, p=1, q=1,
                     par_static=par_stat, coef_est=coef_est)

  sim_data <- sim$simulation$y_sim
}

model <- gas(sim_data, distr=distr, scaling=scaling,
             par_static=par_stat, coef_fix_value=coef_fix_value,
             optim_arguments=list(opts=list(
               algorithm = "NLOPT_LN_NELDERMEAD",
               xtol_rel=1e-05, maxeval=1e+05)))

end_time <- Sys.time()

var_name <- paste("log(", par, ")", sep="")

df[i, "est_omega"] <- model$fit$coef_est[[
  paste(var_name, "_omega", sep="")]]
df[i, "est_score"] <- model$fit$coef_est[[
  paste(var_name, "_alpha1", sep="")]]
df[i, "est_autoreg"] <- model$fit$coef_est[[
  paste(var_name, "_phi1", sep="")]]

if (!is.null(stat_par_name) && !df[i, "stat_par_fix"]) {
  df[i, "est_statpar"] <- model$fit$coef_est[[stat_par_name]]
}

ex_t = as.numeric(end_time - start_time)
print(sprintf("sim: %d/%d time %.2f dist: %s",

```

```
      i, nrow(df), ex_t, distr))

  df[i, "execution_time"] <- ex_t
}

return(df)
}

# example use
sim_df <- setup_gas_sim(N=20, t_sim=c(100),
                        distr = "t", scaling="unit",
                        stat_par1 = 0, stat_par2 = 5,
                        stat_par_fix = c(FALSE, TRUE),
                        true_omega = c(0.2, 0.5, 0.8),
                        true_score = c(0.2, 0.5, 0.8),
                        true_autoreg = c(0.2, 0.5, 0.8))

res_df <- run_gas_sim(sim_df)

saveRDS(res_df, "gas_sim_pois.Rda")
```